My Project

Generated by Doxygen 1.9.1

1 <Game_title> **1**

1.1 HowTo . . . 1

1.1.1 Run game . . . 1

1.1.2 Documentation . . . 1

2 Hierarchical Index **3**

2.1 Class Hierarchy . . . 3

3 Class Index **5**

3.1 Class List . . . 5

4 Class Documentation **7**

4.1 Asteroid Class Reference . . . 7

4.2 Comet Class Reference . . . 7

4.2.1 Constructor & Destructor Documentation . . . 7

4.2.1.1 Comet() . . . 7

4.3 Enemy Class Reference . . . 8

4.4 Exit_State Class Reference . . . 8

4.5 Game_Object Class Reference . . . 9

4.6 Game_State Class Reference . . . 9

4.7 Menu_State Class Reference . . . 10

4.8 Mini Class Reference . . . 10

4.8.1 Constructor & Destructor Documentation . . . 10

4.8.1.1 Mini() . . . 10

4.9 Player Class Reference . . . 11

4.10 Projectile Class Reference . . . 11

4.11 State Class Reference . . . 12

4.12 Texture_Manager Class Reference . . . 12

4.13 Texture_Obj Class Reference . . . 12

4.14 World Class Reference . . . 13

Index **15**

Chapter 1

<Game_title>

1.1 HowTo

1.1.1 Run game

make ./prog

1.1.2 Documentation

```
See file 'class_documentation.pdf' in /docs
```

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

std::enable_shared_from_this
- State . . . 12
  - Exit_State . . . 8
  - Game_State . . . 9
  - Menu_State . . . 10

Game_Object . . . 9
- Texture_Obj . . . 12
  - Enemy . . . 8
    - Asteroid . . . 7
    - Comet . . . 7
    - Mini . . . 10
  - Player . . . 11
  - Projectile . . . 11

Texture_Manager . . . 12

World . . . 13

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Asteroid 7
Comet 7
Enemy 8
Exit_State 8
Game_Object 9
Game_State 9
Menu_State 10
Mini 10
Player 11
Projectile 11
State 12
Texture_Manager 12
Texture_Obj 12
World 13

Chapter 4

Class Documentation

4.1 Asteroid Class Reference

Inheritance diagram for Asteroid:

4.2 Comet Class Reference

Inheritance diagram for Comet:

Collaboration diagram for Comet:

Public Member Functions

- Comet (const sf::Vector2f &pos)

  Enemy type 2: Comet.
- bool **update** (sf::Time delta, World &World) override

Additional Inherited Members

4.2.1 Constructor & Destructor Documentation

4.2.1.1 Comet()

```
Comet::Comet (
            const sf::Vector2f & pos )
```

Enemy type 2: Comet.

Parameters

pos	

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.3 Enemy Class Reference

Inheritance diagram for Enemy:

Collaboration diagram for Enemy:

Public Member Functions

- **Enemy** (const std::string &texture, const float s)

Protected Member Functions

- void **out_of_bounds** ()

Protected Attributes

- const float **speed**
- sf::Vector2f **dir**
- const double **random**

Additional Inherited Members

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.4 Exit_State Class Reference

Inheritance diagram for Exit_State:

Collaboration diagram for Exit_State:

Public Member Functions

- std::shared_ptr< State > **update** (sf::Time)
- void **render** (sf::RenderWindow &)

Additional Inherited Members

The documentation for this class was generated from the following file:

- state.h

4.5 Game_Object Class Reference

Inheritance diagram for Game_Object:

Public Member Functions

- virtual bool **update** (sf::Time time, World &world)=0
- virtual void **render** (sf::RenderWindow &window)=0

Public Attributes

- sf::RectangleShape **shape**

The documentation for this class was generated from the following file:

- game_object.h

4.6 Game_State Class Reference

Inheritance diagram for Game_State:

Collaboration diagram for Game_State:

Public Member Functions

- std::shared_ptr< State > **update** (sf::Time delta) override
- void **render** (sf::RenderWindow &to) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- game_state.h
- game_state.cc

4.7 Menu_State Class Reference

Inheritance diagram for Menu_State:

Collaboration diagram for Menu_State:

Public Member Functions

- **Menu_State** (std::shared_ptr< State > resume=nullptr, int score=0)
- void **on_key_press** (sf::Keyboard::Key key) override
- std::shared_ptr< State > **update** (sf::Time time) override
- void **render** (sf::RenderWindow &draw) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- menu_state.h
- menu_state.cc

4.8 Mini Class Reference

Inheritance diagram for Mini:

Collaboration diagram for Mini:

Public Member Functions

- Mini (const sf::Vector2f &pos)

  Enemy type 1: Asteroid splits into two smaller asteroids (Mini) when hit.
- bool **update** (sf::Time delta, World &World) override

Additional Inherited Members

4.8.1 Constructor & Destructor Documentation

4.8.1.1 Mini()

```
Mini::Mini (
            const sf::Vector2f & pos )
```

Enemy type 1: Asteroid splits into two smaller asteroids (Mini) when hit.

Parameters

pos	

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.9 Player Class Reference

Inheritance diagram for Player:

Collaboration diagram for Player:

Public Member Functions

- **Player** (const sf::Vector2f &pos)
- bool **update** (sf::Time delta, World &world) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- player.h
- player.cc

4.10 Projectile Class Reference

Inheritance diagram for Projectile:

Collaboration diagram for Projectile:

Public Member Functions

- **Projectile** (const sf::Vector2f &pos, const float r)
- bool **update** (sf::Time delta, World &world) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- projectile.h
- projectile.cc

4.11 State Class Reference

Inheritance diagram for State:

Collaboration diagram for State:

Public Member Functions

- virtual void **on_key_press** (sf::Keyboard::Key key)
- virtual void **on_key_release** (sf::Keyboard::Key key)
- virtual std::shared_ptr< State > **update** (sf::Time time)=0
- virtual void **render** (sf::RenderWindow &to)=0

Static Public Member Functions

- static void **run** (sf::RenderWindow &window, std::shared_ptr< State > state)

The documentation for this class was generated from the following files:

- state.h
- state.cc

4.12 Texture_Manager Class Reference

Static Public Member Functions

- static sf::Texture * **get** (const std::string &name)

The documentation for this class was generated from the following files:

- texture_manager.h
- texture_manager.cc

4.13 Texture_Obj Class Reference

Inheritance diagram for Texture_Obj:

Collaboration diagram for Texture_Obj:

Public Member Functions

- **Texture_Obj** (const std::string &texture)
- void **render** (sf::RenderWindow &window) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- game_object.h
- game_object.cc

4.14 World Class Reference

Public Member Functions

- void **update** (sf::Time delta)
- void **render** (sf::RenderWindow &to)
- void **add** (shared_ptr< Game_Object > object)
- std::vector< shared_ptr< Game_Object > > **collides_with** (Game_Object &me) const
- bool **is_dead** () const

Public Attributes

- int **score** {0}
- int **maxEnemies** {0}

The documentation for this class was generated from the following files:

- world.h
- world.cc

Index

Asteroid, 7

Comet, 7
    Comet, 7

Enemy, 8
Exit_State, 8

Game_Object, 9
Game_State, 9

Menu_State, 10
Mini, 10
    Mini, 10

Player, 11
Projectile, 11

State, 12

Texture_Manager, 12
Texture_Obj, 12

World, 13